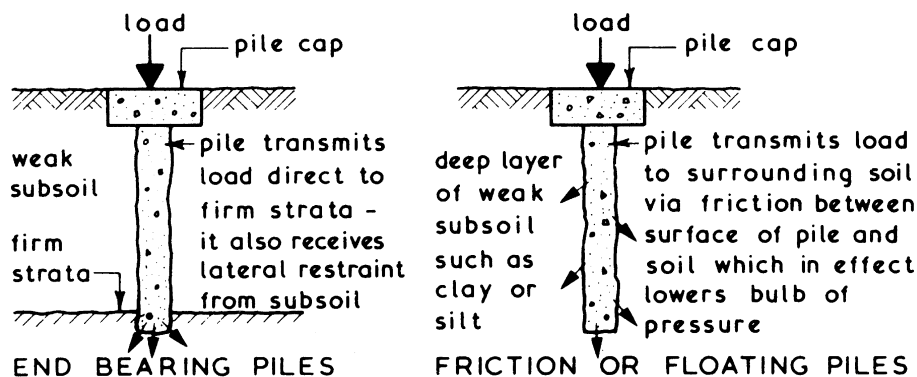


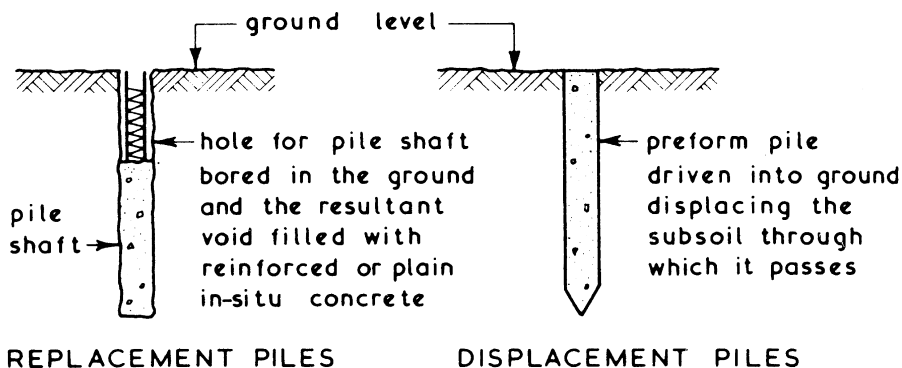
Piled Foundations ~ these can be defined as a series of columns constructed or inserted into the ground to transmit the load(s) of a structure to a lower level of subsoil. Piled foundations can be used when suitable foundation conditions are not present at or near ground level making the use of deep traditional foundations uneconomic. The lack of suitable foundation conditions may be caused by:

1. Natural low bearing capacity of subsoil.
2. High water table - giving rise to high permanent dewatering costs.
3. Presence of layers of highly compressible subsoils such as peat and recently placed filling materials which have not sufficiently consolidated.
4. Subsoils which may be subject to moisture movement or plastic failure.

Classification of Piles ~ piles may be classified by their basic design function or by their method of construction:

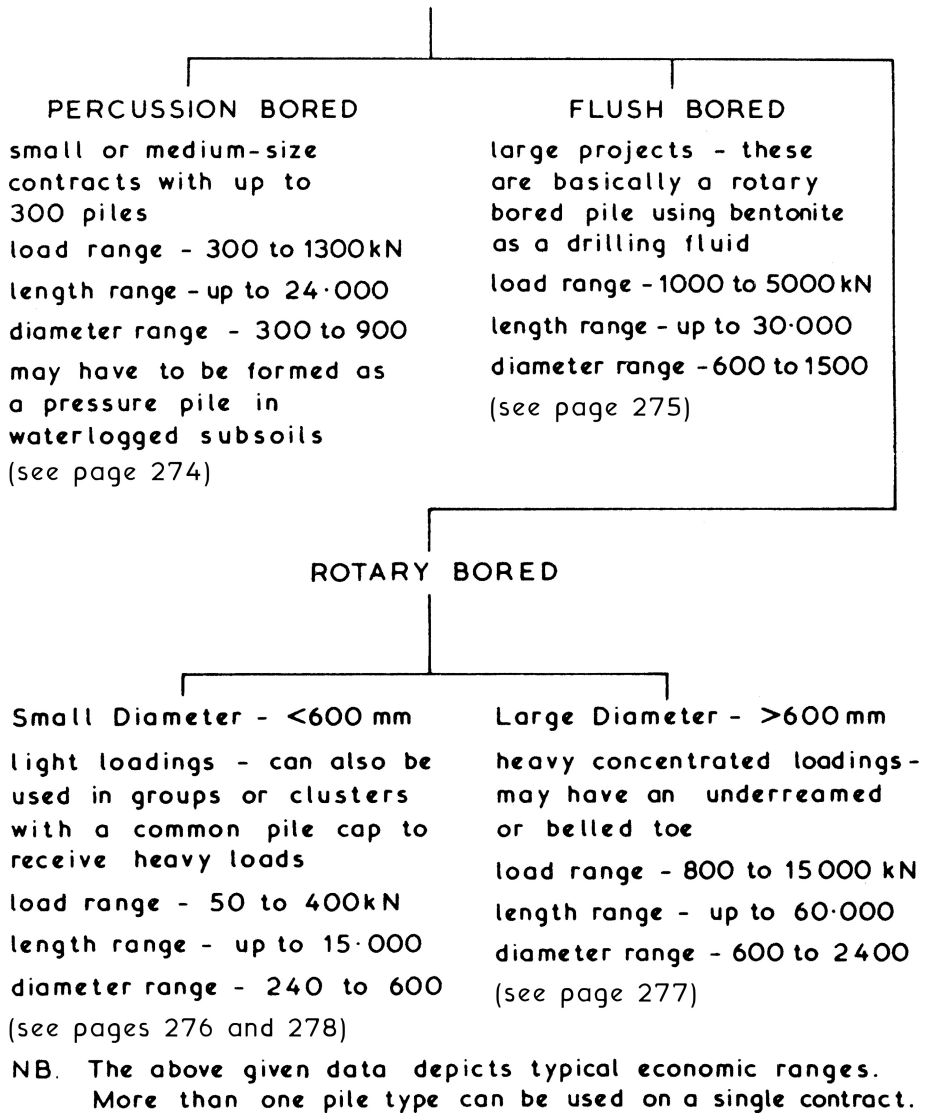


NB. Piles can work in a combination of the above design functions



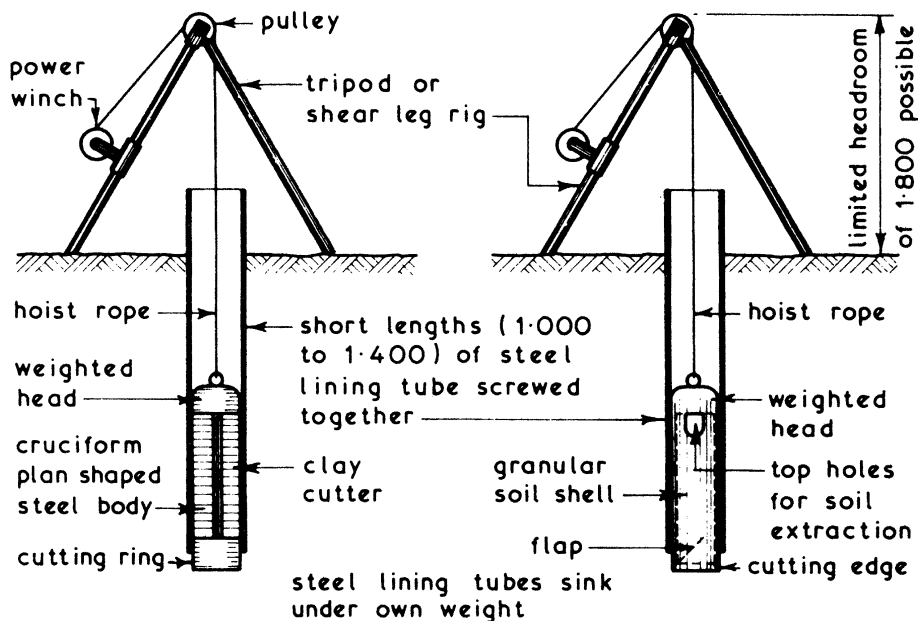
Replacement Piles ~ these are often called bored piles since the removal of the spoil to form the hole for the pile is always carried out by a boring technique. They are used primarily in cohesive subsoils for the formation of friction piles and when forming pile foundations close to existing buildings where the allowable amount of noise and/or vibration is limited.

Replacement Pile Types ~



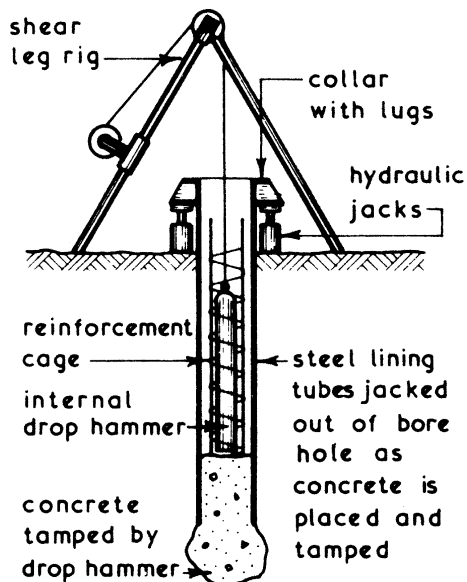
Percussion Bored Piles

Typical Details ~

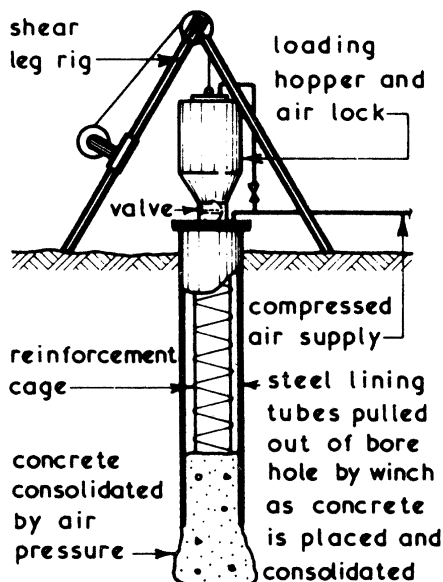


BORING IN COHESIVE SOIL

BORING IN GRANULAR SOIL



FORMING PILE USING DROP HAMMER

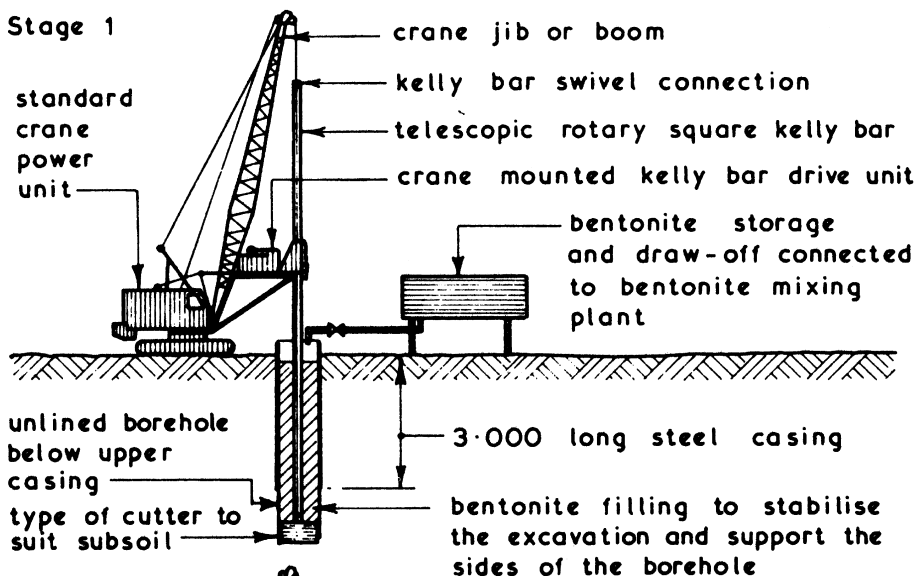


FORMING PILE USING AIR PRESSURE

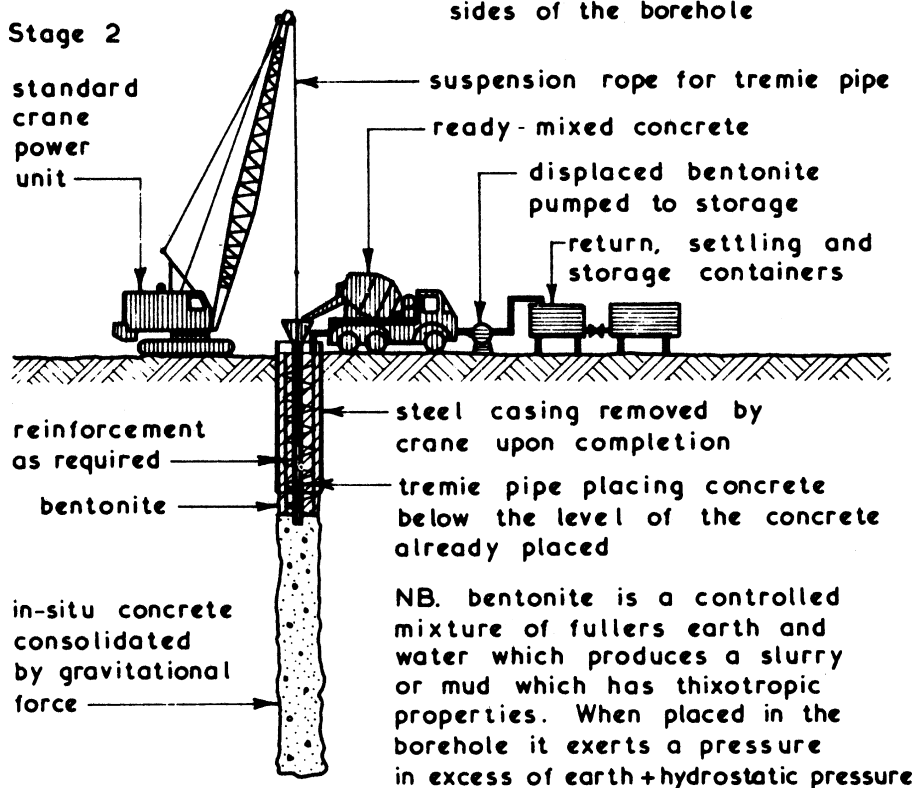
Flush Bored Piles

Typical Details ~

Stage 1

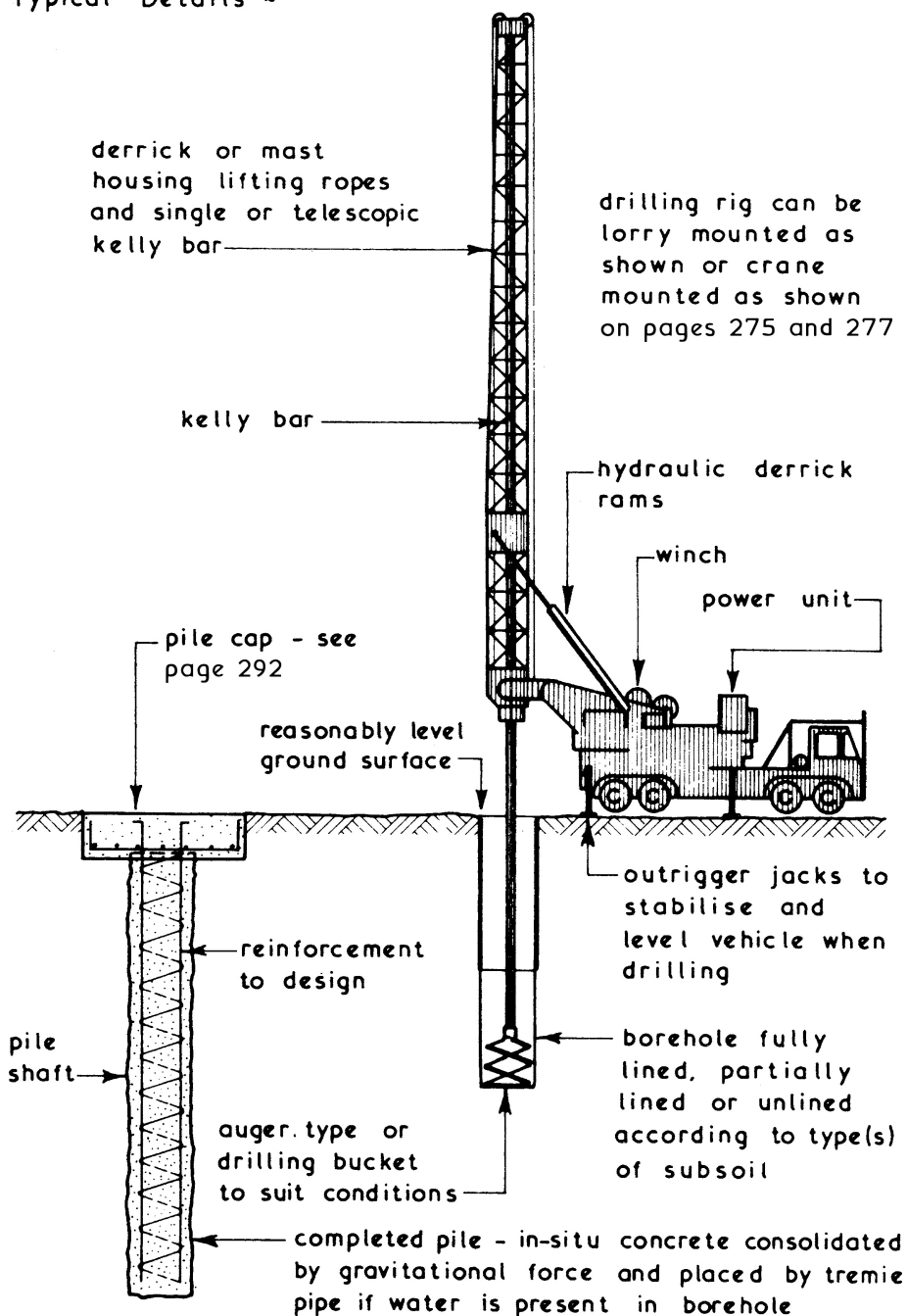


Stage 2



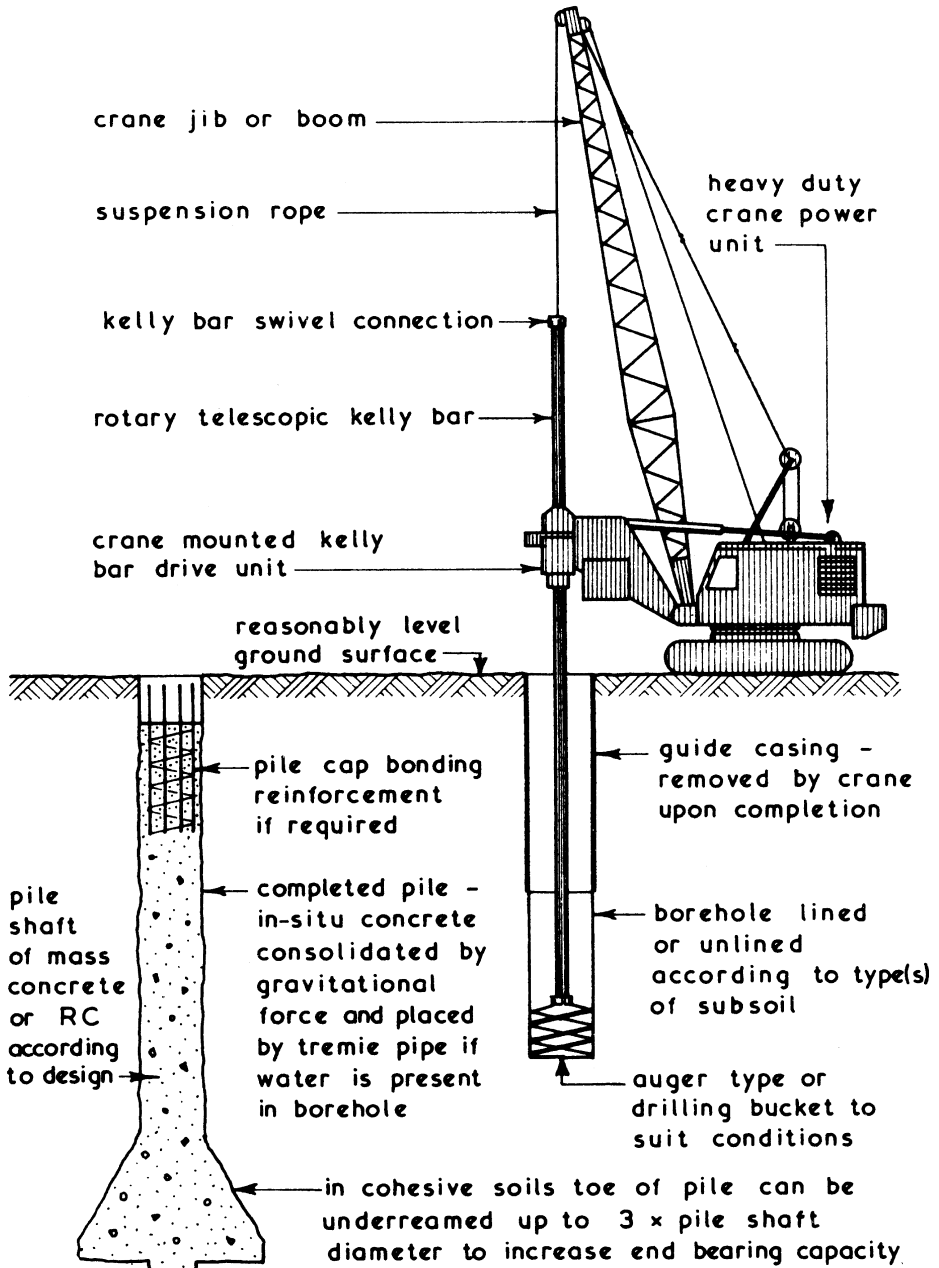
Small Diameter Rotary Bored Piles

Typical Details ~



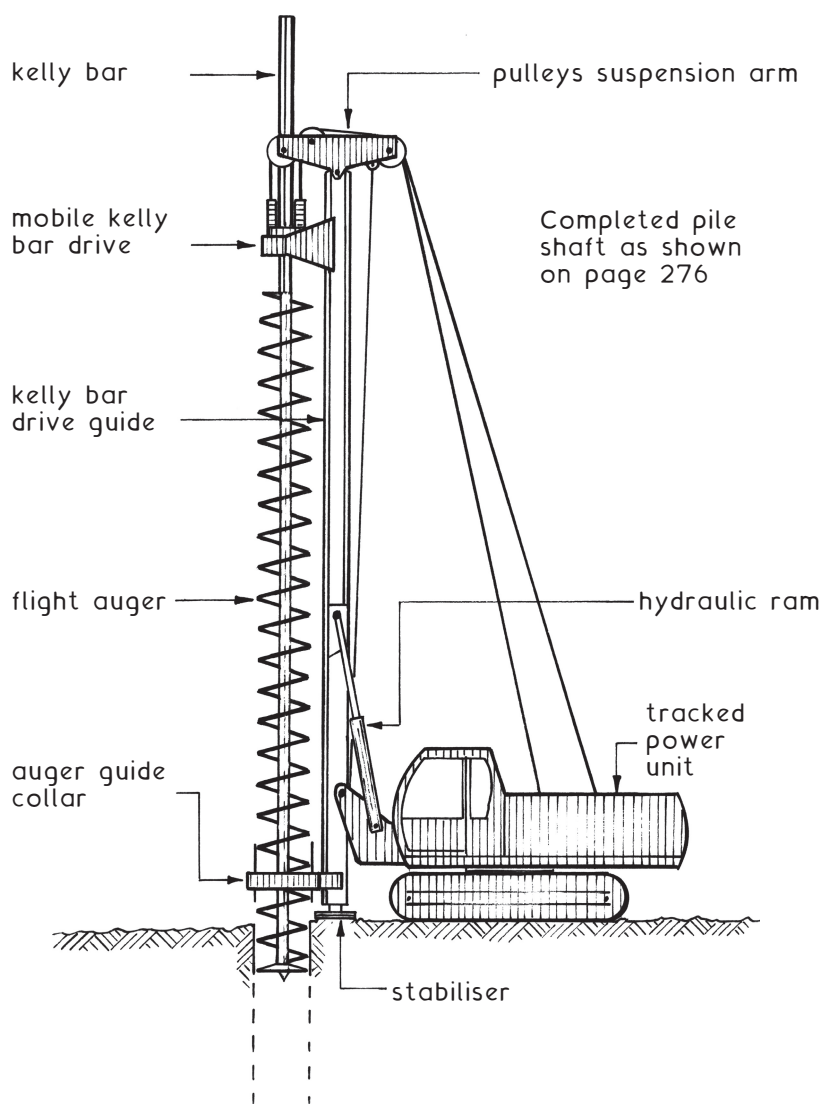
Large Diameter Rotary Bored Piles

Typical Details ~



Continuous Flight Auger Bored Piles

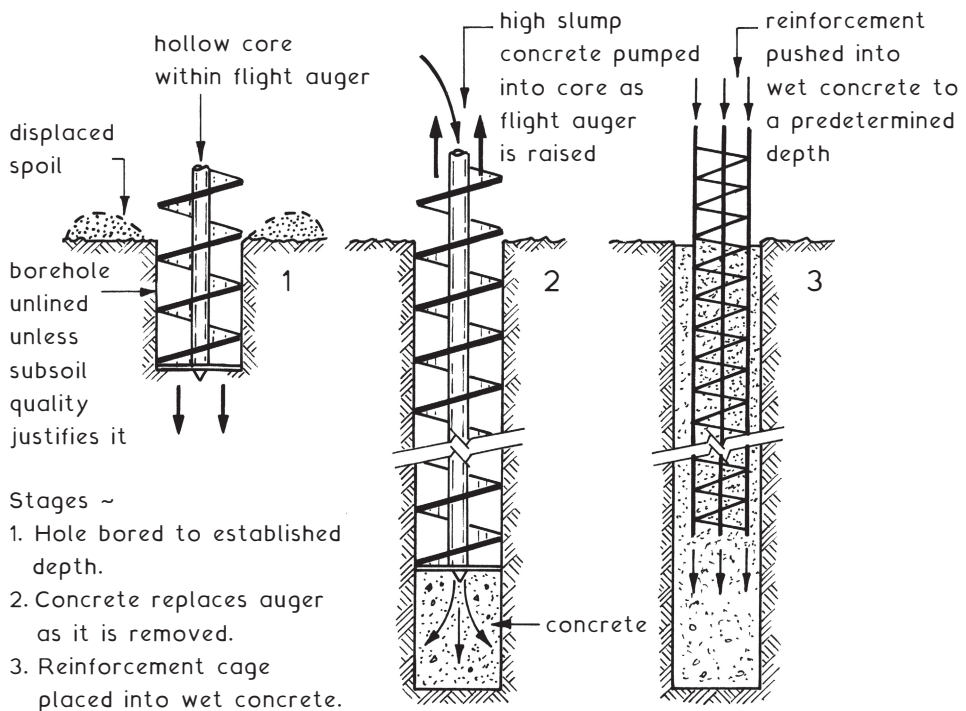
Typical Details ~



Standard pile diameters are 300, 450 and 600 mm. Depth of borehole to about 15 m.

Grout Injection Piling ~

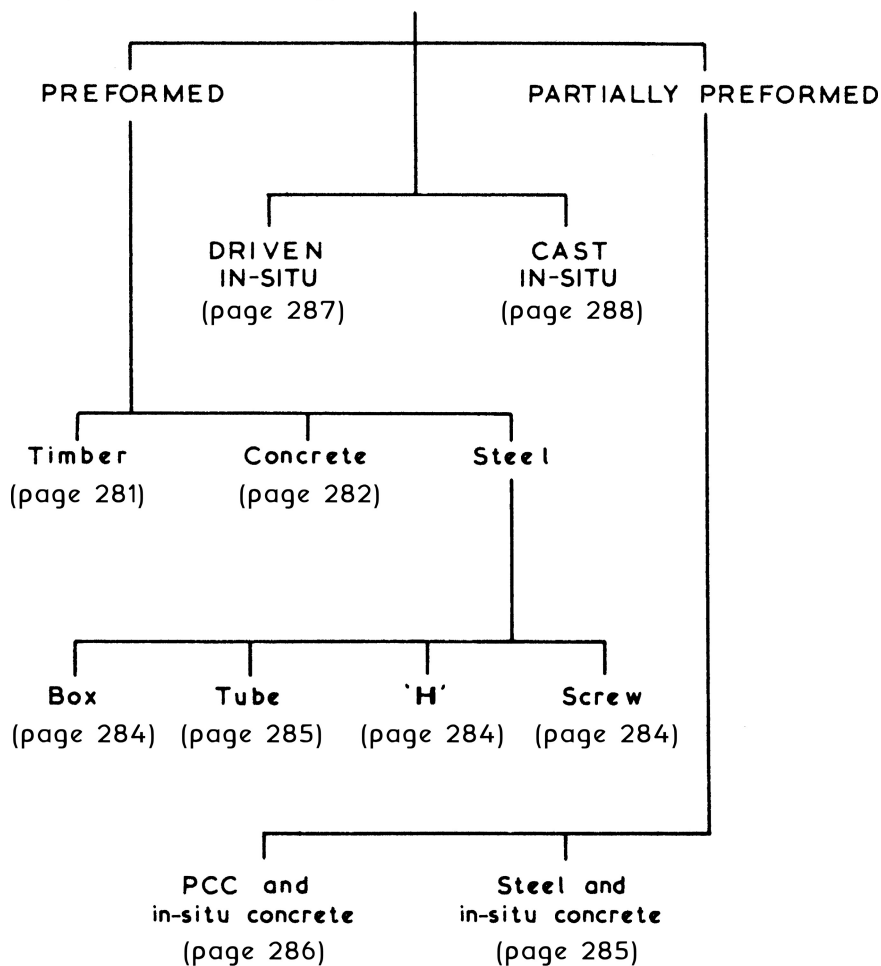
A variation of continuous flight auger bored piling that uses an open ended hollow core to the flight. After boring to the required depth, high slump concrete is pumped through the hollow stem as the auger is retracted. Spoil is displaced at the surface and removed manually. In most applications there is no need to line the boreholes, as the subsoil has little time to be disturbed. A preformed reinforcement cage is pushed into the wet concrete.



Piled Foundations

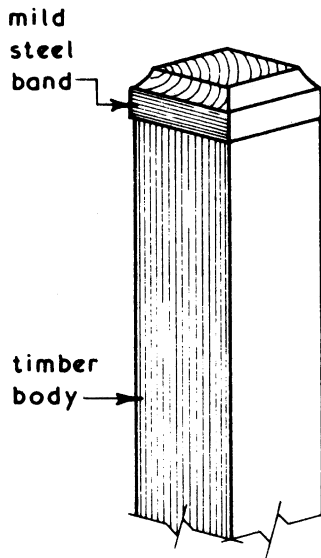
Displacement Piles ~ these are often called driven piles since they are usually driven into the ground, displacing the earth around the pile shaft. These piles can be either preformed or partially preformed if they are not cast in-situ and are available in a wide variety of types and materials. The pile or forming tube is driven into the required position to a predetermined depth or to the required 'set' which is a measure of the subsoil's resistance to the penetration of the pile and hence its bearing capacity by noting the amount of penetration obtained by a fixed number of hammer blows.

Displacement Pile Types ~



Timber Piles ~ these are usually square sawn and can be used for small contracts on sites with shallow alluvial deposits overlaying a suitable bearing strata (e.g. river banks and estuaries). Timber piles are percussion driven.

Typical Example ~

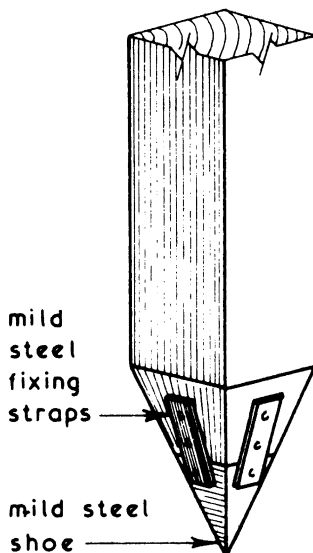


Typical Data:

load range - 50 to 350kN
length range - up to 12·000
without splicing

size range - 225 × 225
300 × 300 *
350 × 350 *
400 × 400 *
450 × 450
600 × 600

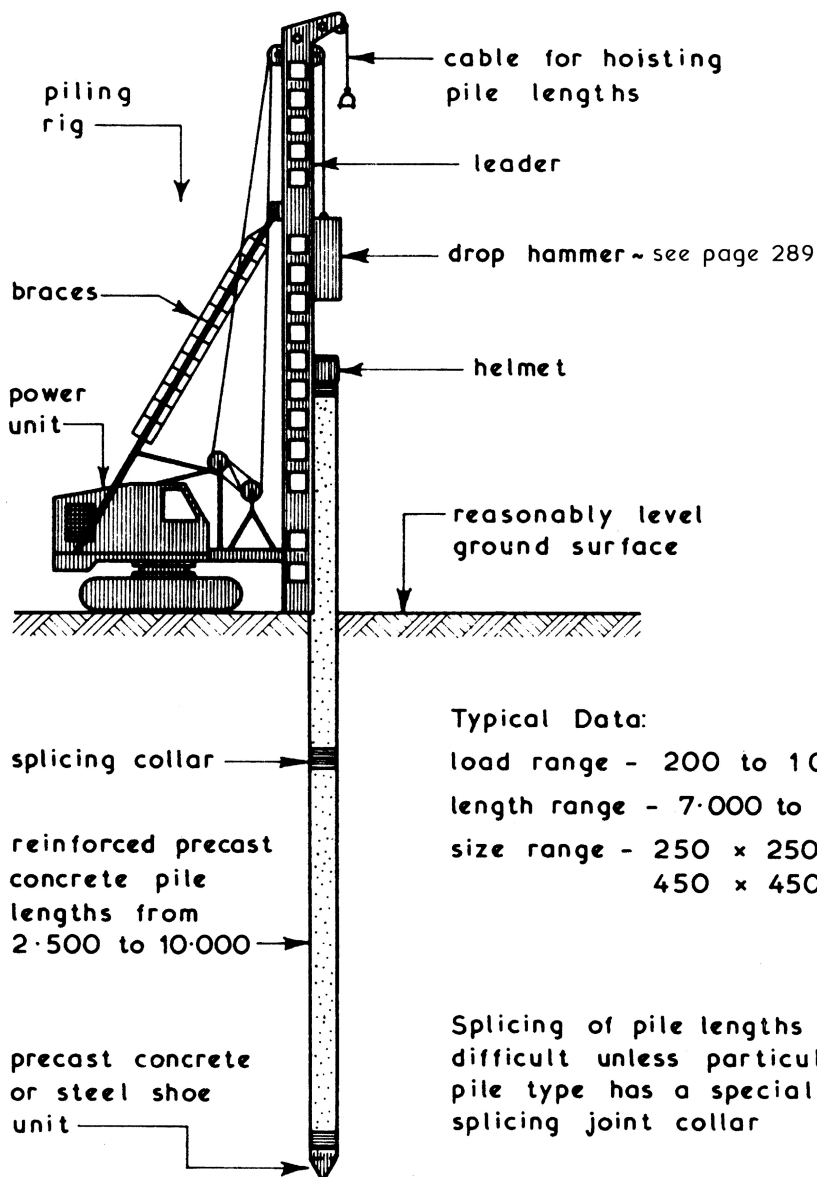
* common sizes



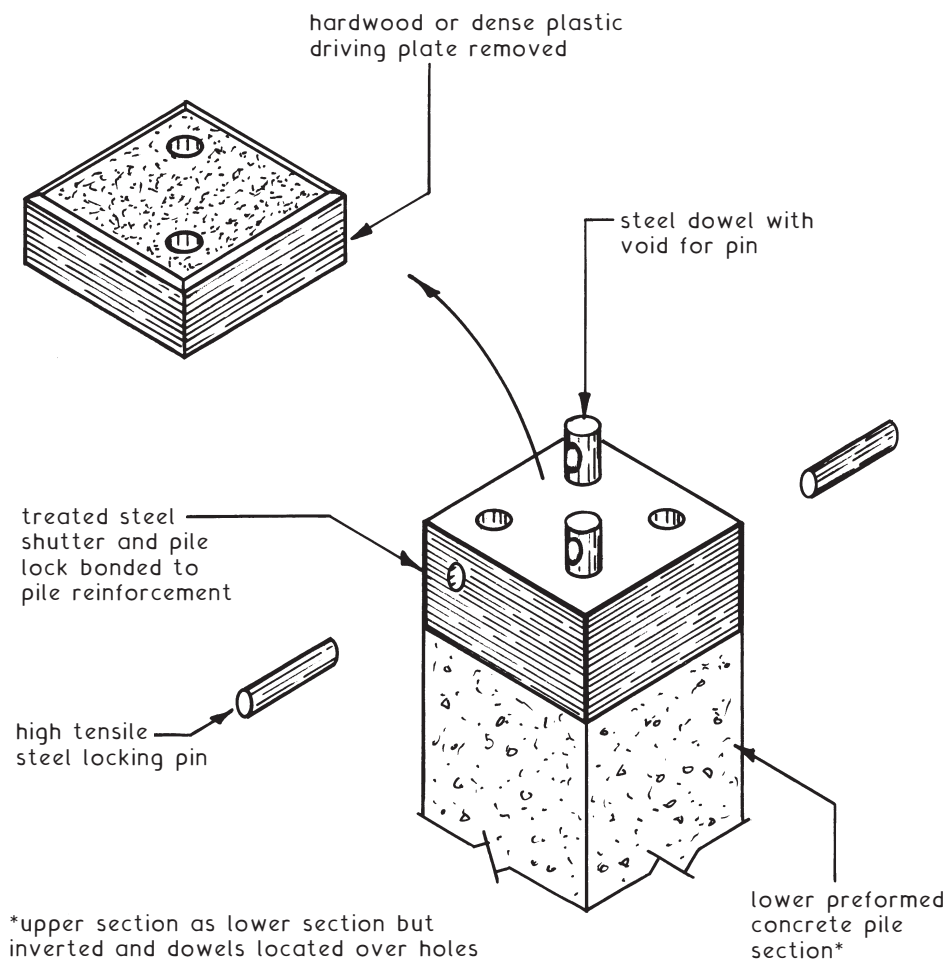
NB. timber piles are not easy to splice and are liable to attack by marine borers when set in waters; therefore such piles should always be treated with a suitable preservative before being driven.

Preformed Concrete Piles ~ variety of types available which are generally used on medium to large contracts of not less than one hundred piles where soft soil deposits overlie a firmer strata. These piles are percussion driven using a drop or single-acting hammer.

Typical Example [West's Hardrive Precast Modular Pile] ~



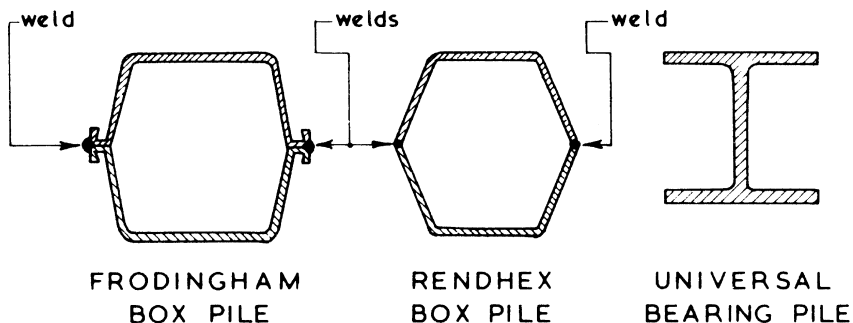
Preformed Concrete Piles – jointing with a peripheral steel splicing collar as shown on the preceding page is adequate for most concentrically or directly loaded situations. Where very long piles are to be used and/or high stresses due to compression, tension and bending from the superstructure or the ground conditions are anticipated, the four or eight lock pile joint [AARSLEFF PILING] may be considered.



Pile dimensions (mm)	Possible no. of locks per joint
250 × 250, 300 × 300, 350 × 350 and 400 × 400	4
350 × 350, 400 × 400 and 450 × 450	8

Steel Box and 'H' Sections ~ standard steel sheet pile sections can be used to form box section piles whereas the 'H' section piles are cut from standard rolled sections. These piles are percussion driven and are used mainly in connection with marine structures.

Typical Examples ~

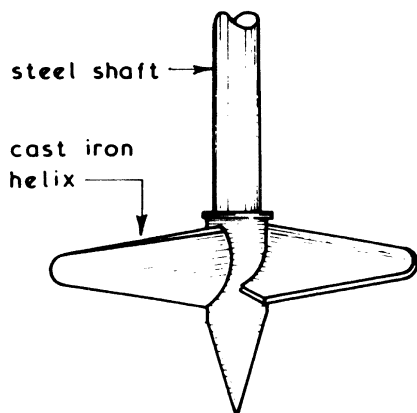


Typical Data:

load range - box piles 300 to 1500 kN
 bearing piles 300 to 1700 kN
 length range - all types up to 36·000
 size range - various sizes and profiles available

Steel Screw Piles ~ rotary driven and used for dock and jetty works where support at shallow depths in soft silts and sands is required.

Typical Example ~

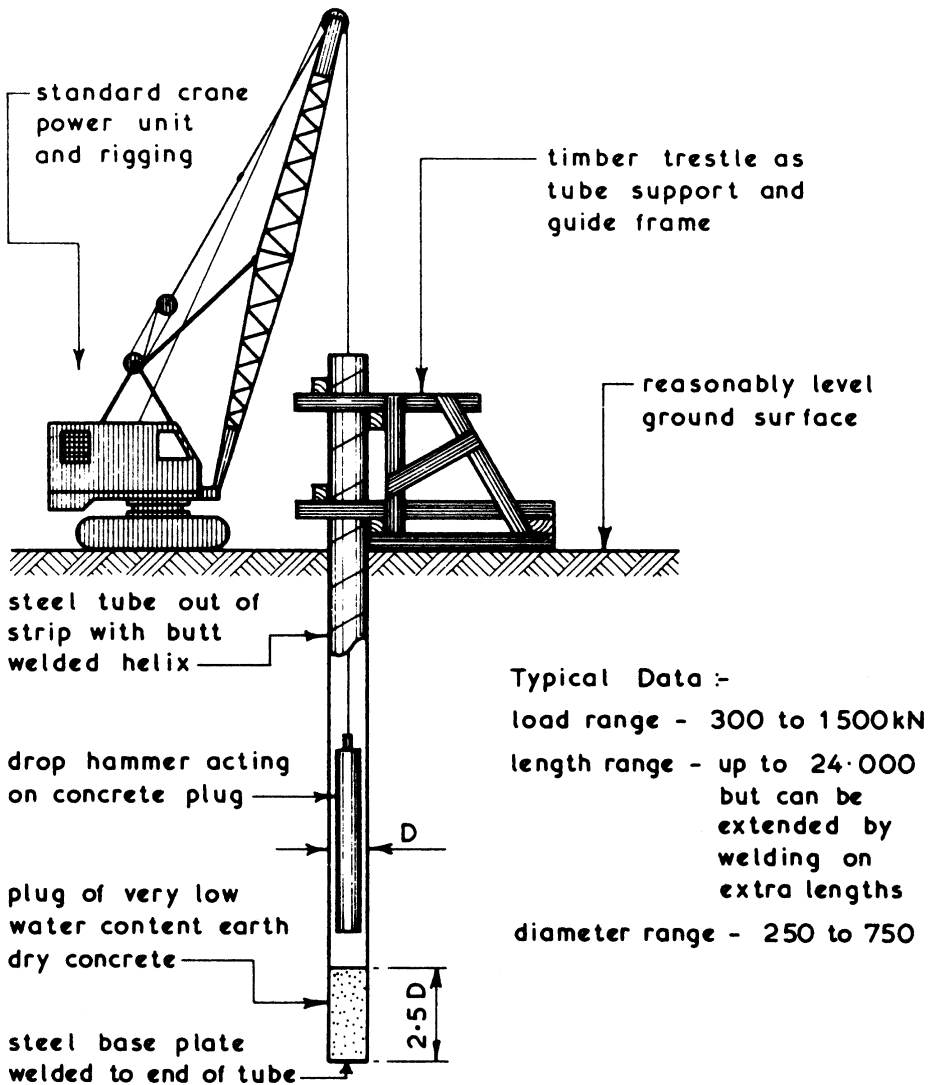


Typical Data:

load range - 400 to 3000 kN
 length range - up to 24·000
 size range - shafts 150 to 350mm dia.
 overall blades 600 to 1200

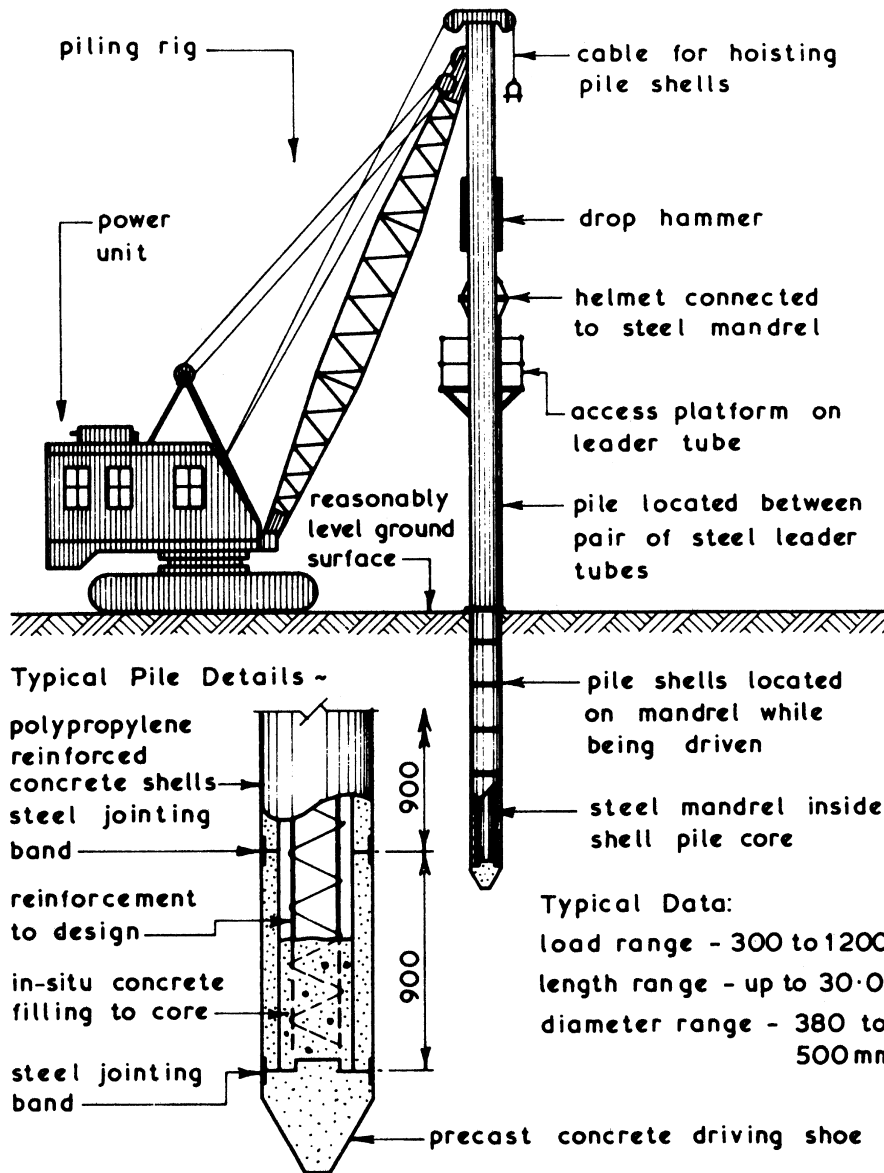
Steel Tube Piles ~ used on small to medium-size contracts for marine structures and foundations in soft subsoils over a suitable bearing strata. Tube piles are usually bottom driven with an internal drop hammer. The loading can be carried by the tube alone but it is usual to fill the tube with mass concrete to form a composite pile. Reinforcement, except for pile cap bonding bars, is not normally required.

Typical Example [BSP Cased Pile] ~



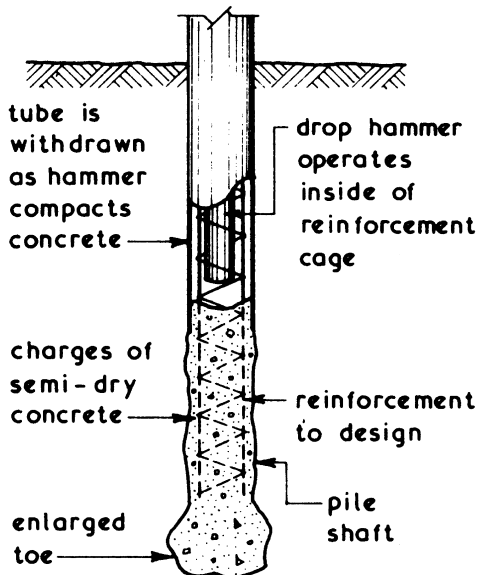
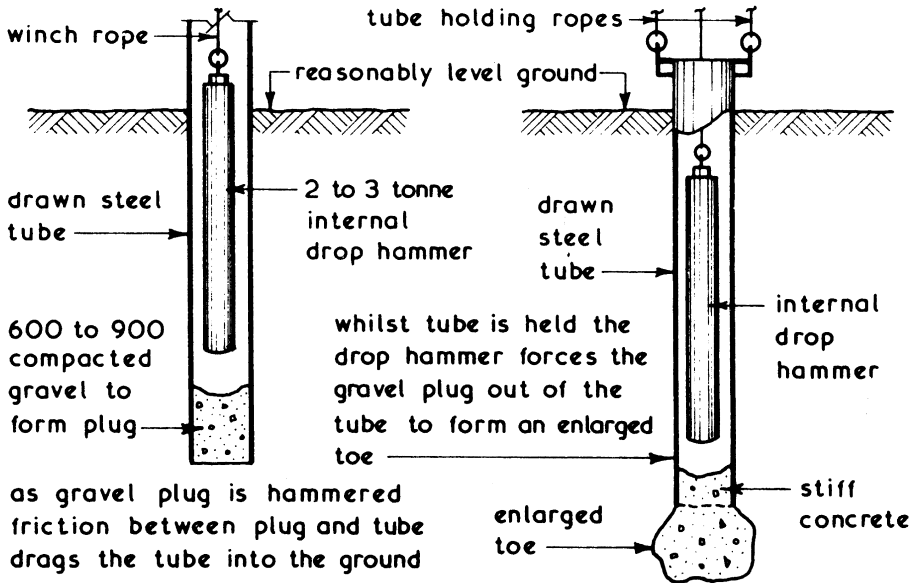
Partially Preformed Piles ~ these are composite piles of precast concrete and in-situ concrete or steel and in-situ concrete (see page 285). These percussion-driven piles are used on medium to large contracts where bored piles would not be suitable owing to running water or very loose soils.

Typical Example [West's Shell Pile] ~



Driven In-situ Piles ~ used on medium to large contracts as an alternative to preformed piles particularly where final length of pile is a variable to be determined on site.

Typical Example [Franki Driven In-situ Pile] ~



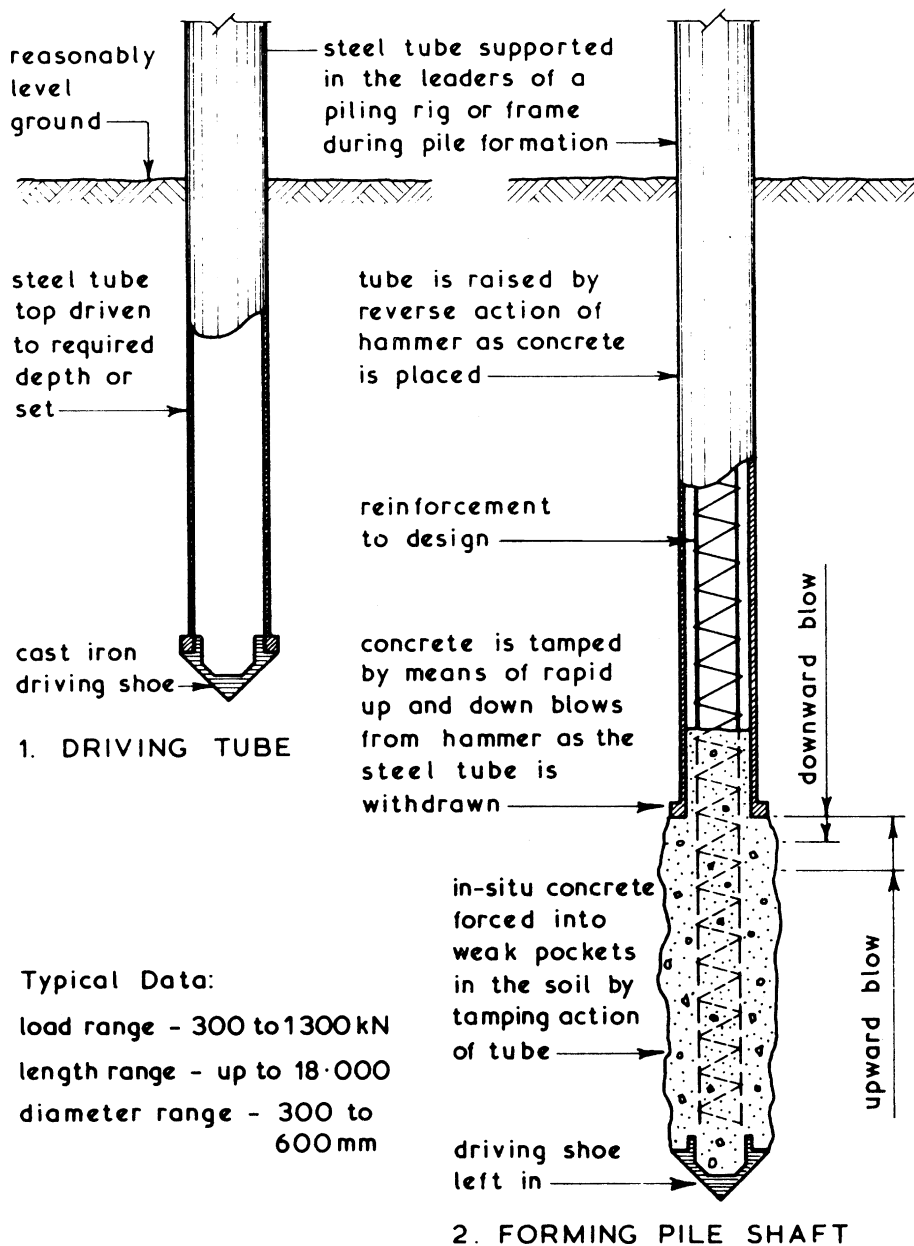
The drawn steel tube is supported in the leaders of a piling rig or frame during the formation of the pile

Typical Data:

load range - 300 to 1300 kN
 length range - up to 18.000
 diameter range - 300 to 600 mm

Cast In-situ Piles ~ an alternative to the driven in-situ piles (see page 287)

Typical Example [Vibro Cast In-situ Pile] ~



Typical Data:

load range - 300 to 1300 kN

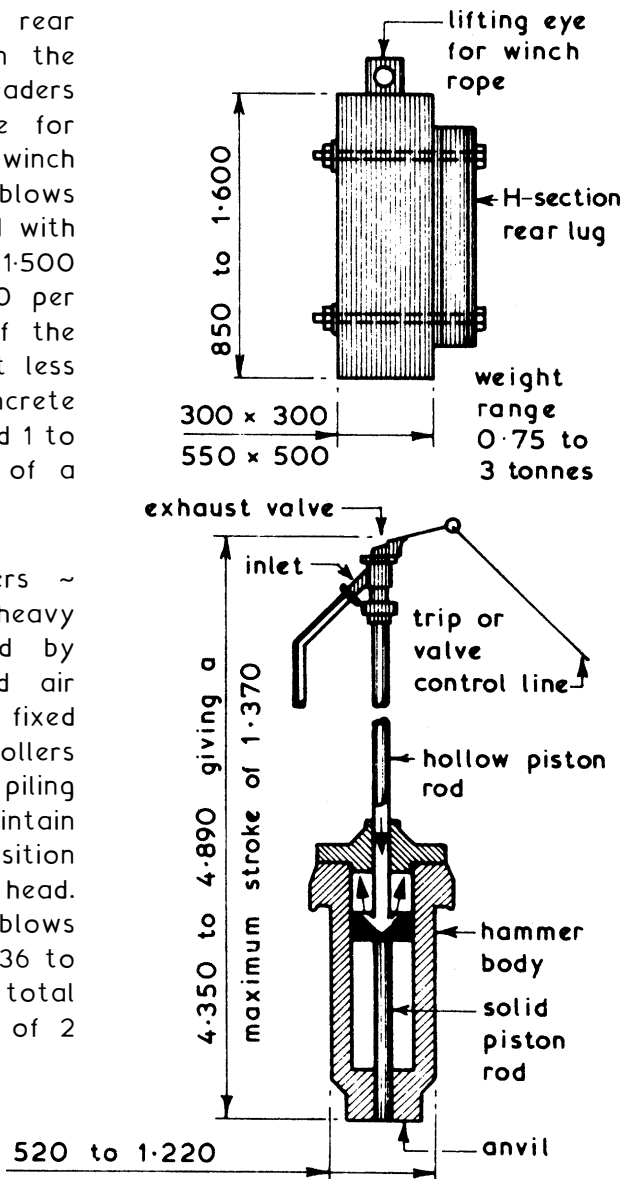
length range - up to 18.000

diameter range - 300 to 600 mm

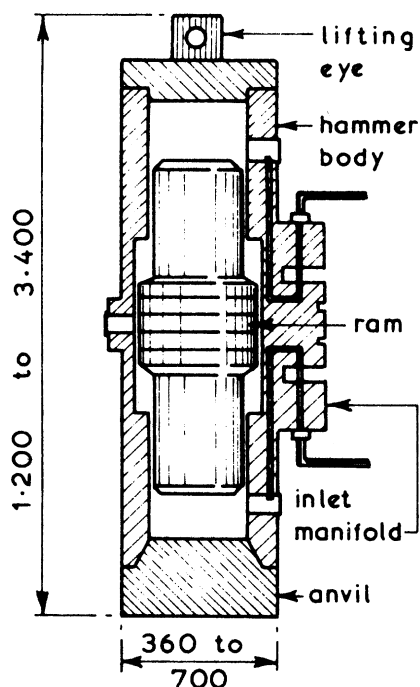
Piling Hammers ~ these are designed to deliver an impact blow to the top of the pile to be driven. The hammer weight and drop height is chosen to suit the pile type and nature of subsoil(s) through which it will be driven. The head of the pile being driven is protected against damage with a steel helmet which is padded with a sand bed or similar material and is cushioned with a plastic or hardwood block called a dolly.

Drop Hammers ~ these are blocks of iron with a rear lug(s) which locate in the piling rig guides or leaders and have a top eye for attachment of the winch rope. The number of blows which can be delivered with a free fall of 1.200 to 1.500 ranges from 10 to 20 per minute. The weight of the hammer should be not less than 50% of the concrete or steel pile weight and 1 to 1.5 times the weight of a timber pile.

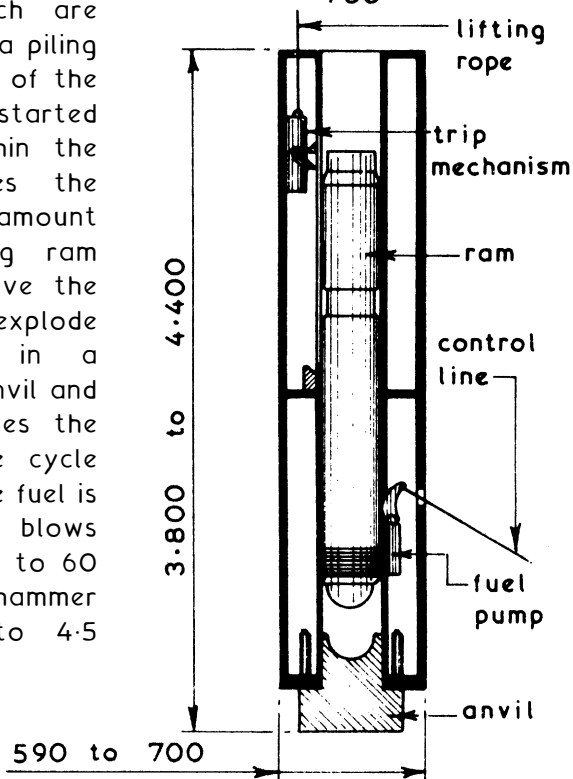
Single Acting Hammers ~ these consist of a heavy falling cylinder raised by steam or compressed air sliding up and down a fixed piston. Guide lugs or rollers are located in the piling frame leaders to maintain the hammer position relative to the pile head. The number of blows delivered ranges from 36 to 75 per minute with a total hammer weight range of 2 to 15 tonnes.



Double Acting Hammers ~ these consist of a cast iron cylinder which remains stationary on the pile head whilst a ram powered by steam or compressed air for both up and down strokes delivers a series of rapid blows which tends to keep the pile on the move during driving. The blow delivered is a smaller force than that from a drop or single acting hammer. The number of blows delivered ranges from 95 to 300 per minute with a total hammer weight range of 0.7 to 6.5 tonnes. Diesel powered double acting hammers are also available.



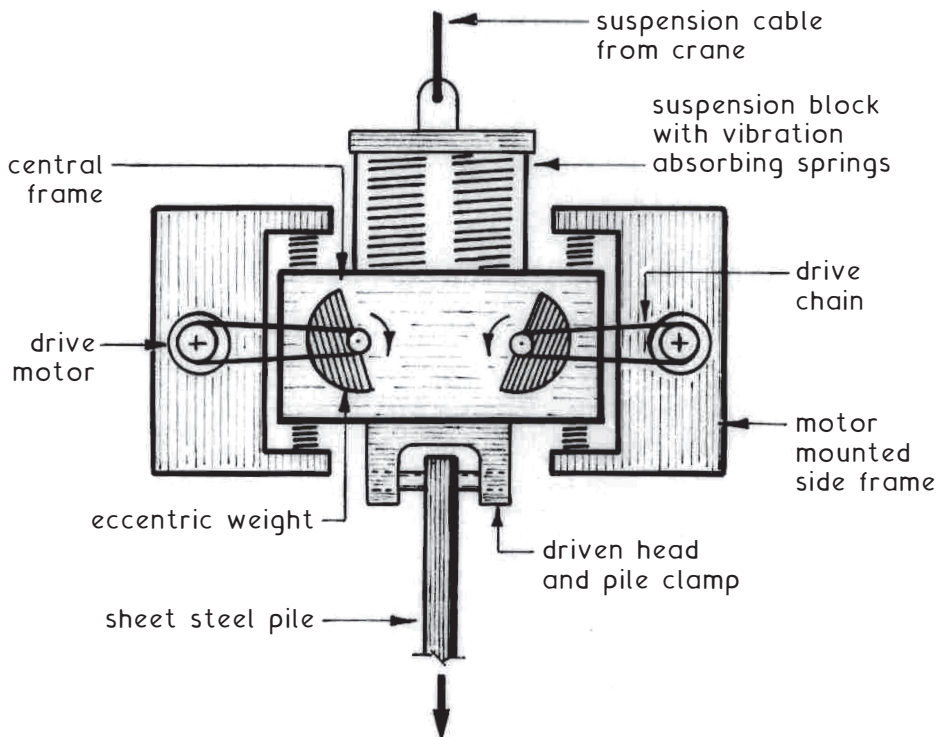
Diesel Hammers ~ these are self-contained hammers which are located in the leaders of a piling rig and rest on the head of the pile. The driving action is started by raising the ram within the cylinder which activates the injection of a measured amount of fuel. The free-falling ram compresses the fuel above the anvil causing the fuel to explode and expand resulting in a downward force on the anvil and upward force which raises the ram to recommence the cycle which is repeated until the fuel is cut off. The number of blows delivered ranges from 40 to 60 per minute with a total hammer weight range of 1.0 to 4.5 tonnes.



Vibrating Hammer ~ used with preformed displacement piles driven into clay, sand or gravel subsoils.

Application ~ unit suspended by cable from a crane, positioned and secured to the top of the pile. Steel pipe, H section and sheet pile profiles are most suited to this procedure.

Function ~ two electric motors are used to propel eccentric rotors in opposing directions. The vibrations generated combine with the weight of the unit to transmit through the pile. This breaks down the subsoils resistance to shear and enables the pile to penetrate. Resistance can be reduced by water jetting at the toe of the pile.



Advantages ~ relatively silent in operation, a considerable benefit in built-up areas. Use of electricity is non-polluting in the immediate site area. No impact, therefore pile cap is unlikely to be damaged.

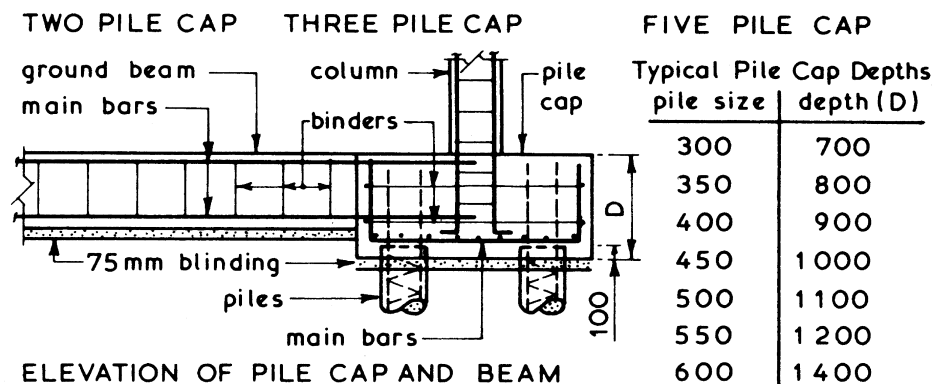
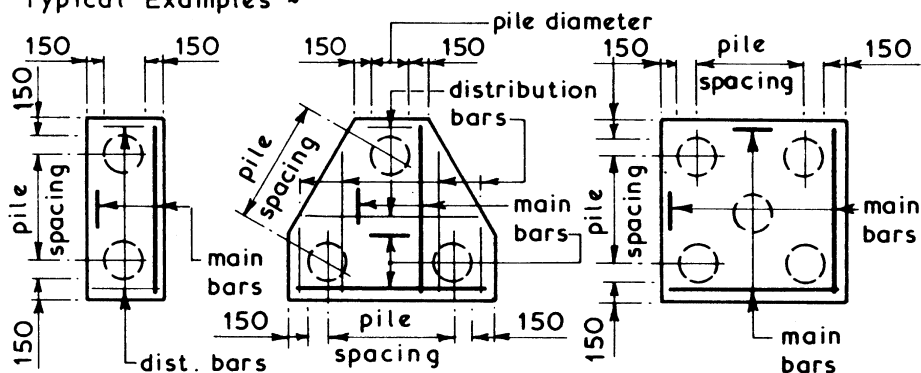
Disadvantage ~ electricity is not always available during preliminary ground works.

Pile Caps ~ piles can be used singly to support the load but often it is more economical to use piles in groups or clusters linked together with a reinforced concrete cap. The pile caps can also be linked together with reinforced concrete ground beams.

The usual minimum spacing for piles is:

1. Friction Piles - $1:100$ or not less than $3 \times$ pile diameter, whichever is the greater.
2. Bearing Piles - 750 mm or not less than $2 \times$ pile diameter, whichever is the greater.

Typical Examples ~



Pile Testing ~ it is advisable to test load at least one pile per scheme. The test pile should be overloaded by at least 50% of its working load and this load should be held for 24 hours. The test pile should not form part of the actual foundations. Suitable testing methods are:

1. Jacking against kentledge placed over test pile.
2. Jacking against a beam fixed to anchor piles driven in on two sides of the test pile.